Gregory Bian

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Education

University of British Columbia Expected May 2028
Bachelor of Applied Science, Computer Engineering *Vancouver, BC*

Technical Experience

Geotab Burnaby, BC
Embedded Software Engineer Intern – Vehicle Platform Development *May 2026 – Present*

- Developed production code on 6M+ vehicles, adding 15+ OEM vehicle types to Geotab's telematics platform
- Collected 10B+ data points by designing signal processing work flows with dependency gating, software filtering and automated validation to ensure accurate vehicle data collection and backend processing
- Migrated 200 SIL test suites to use real, unmasked vehicle data for VIN decoding and vehicle identification, improving test coverage and removing the need for bench simulations
- Reduced stale feature flag removal effort by 90% by automating code edits, PRs, Jira updates, and CI/SIL monitoring end-to-end with Claude Skills on VMs; removed 50+ flags and adopted team-wide.

UBC Solar Racing Vancouver, BC
Embedded Systems Team Lead *Jun 2026 – Present*

- Led a 9-person subteam within an 80+ member team, delivering firmware and telemetry for a 4-ECU solar car
- Planned project milestones for a 2-year design cycle, directing design and code reviews and mentoring members
- Architected SIL and Regression testing pipelines using Renode hardware emulation and Robot testing Frameworks to simulate RTOS and hardware edge cases

Senior Designer Engineer *Apr 2026 – Jun 2026*

- Validated safety-critical systems before race day by leading end-to-end integration tests of radio/cellular telemetry, motor control safety features, and BMS interfaces
- Developed automated Bash and Python testing scripts to improve the reliability and coverage of integration tests

Embedded Firmware Engineer *Sep 2025 – Apr 2026*

- Developed firmware to display 6x more data on the LCD Dashboard than previous iterations by architecting a multi-page design, improving real-time driver awareness, fault detection, and decision making
- Characterized and eliminated live telemetry packet loss (250 bytes/s $\rightarrow$ 0) by using RTOS semaphores in UART drivers for an XBee radio module to synchronize shared buffers across concurrent tasks
- Added 50+ vehicle signals to the CAN bus including motor temperature, GPS coordinates, and various diagnostics
- Diagnosed 5+ hardware issues on custom PCBs by probing signals with oscilloscopes, multimeters, and PCANs

Projects

NotiFlow Course Tracker | *JavaScript (React), Flask, Python* Nov 2025 – Dec 2025

- Led a team of 5 in developing a Course Tracking Chrome Extension; implemented a React-based UI and integrated Chrome Local Storage APIs to store data locally, improving usability and reliability.
- Leveraged Flask routing to send and receive data between the Chrome Extension and backend Python scripts and support features such as Canvas API access, BeautifulSoup webscraping and OpenAI API prompt filtering.

Extracurriculars

nwPlus Vancouver, BC
Engagement Coordinator *May 2026 – Present*

- Organized the largest hackathon in western Canada, hosting 750+ students, 50+ sponsors, and 150+ mentors
- Fostered a collaborative community within a 50+ member organizing committee by planning team-building events

Technical Skills

Programming: C, C++, Python, Java, JavaScript (React), Bash, ARM Assembly
Embedded Systems: FreeRTOS, STM32, CAN (OBD-II, UDS, ISO-TP, J1939), UART, SPI, I2C
Tools: Claude Code, Linux (Ubuntu), Git, J-Link, STM32CubeMX, CMake, Grafana, InfluxDB, SavvyCAN